



May 2025

(For CV)

Max. Marks: 100

Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12}$ F/m, Permeability of free space, $\mu_0 = 4\pi \times 10^{-7}$ H/m

20 Marks

- 1-



11. Nd:YAG laser is a
 - A) Two level laser
 - B) Three level laser
 - C) Four level laser
 - D) Five level laser
12. Laser action is found in _____ semiconductor.
 - A) direct band gap
 - B) indirect band gap
 - C) germanium
 - D) silicon
13. Optical fiber works on the phenomenon of _____.
 - A) total internal reflection
 - B) polarization
 - C) diffraction
 - D) refraction
14. What causes microscopic bend in a optical fiber?
 - A) Absorption
 - B) Scattering
 - C) Impurities
 - D) Non-uniform pressure
15. The loss in signal power as light travels down a fiber is called
 - A) Dispersion
 - B) Attenuation
 - C) Scattering
 - D) Absorption
16. Signal distortion occurs due to
 - A) Irregularities in fiber structure
 - B) Variation in refractive index of the core at different points
 - C) Spreading of pulse
 - D) Macroscopic bend
17. The sound waves having frequencies greater than 20 KHz are called as waves.
 - A) Ultrasonic
 - B) Infrasonic
 - C) Audible
 - D) Supersonic
18. According to Weber- Fechner law, Loudness is proportional to..... where I is the intensity of sound.
 - A) Square root of I
 - B) $\ln I$
 - C) $\log I$
 - D) $1/\log I$
19. Which characteristic of sound is primarily associated with the pitch of a musical note?
 - A) Amplitude
 - B) Frequency
 - C) Wavelength
 - D) Speed
20. Which of the following units is used to measure loudness?
 - A) Decibels
 - B) Hertz
 - C) Watts
 - D) Meters

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is simple harmonic motion? Derive the differential equation for simple harmonic motion.	4	L*2	1	1, 2
	b) Derive the expression for effective spring constant of springs in series and parallel.	8	L2	1	1, 2
	c) A cube of aluminium oxide of side 0.1 m is subjected to shearing force of 100 N. The top surface of the cube is displaced by 0.01 cm with respect to bottom. Calculate the shearing stress, shearing strain and modulus of rigidity.	4	L3	1	1, 2
2.	a) Explain Free, Damped and Forced oscillations.	6	L2	1	1, 2
	b) Derive the expression for twisting couple of a wire.	6	L2	2	1, 2
	c) A block of mass is 680 g is suspended to a string of spring constant 65 N/m. The block is made to oscillate. Calculate the angular frequency, frequency and period of oscillation.	4	L3	1	1, 2
3.	a) Define Poisson's ratio. Derive the relation connecting the three elastic moduli.	6	L2	2	1, 2
	b) What is bending moment? Derive an expression for the bending moment of a beam.	6	L1, L2	2	1, 2
	c) What is the force required to stretch the steel wire of cross section 0.5 cm ² to double its length? Given Young's modulus 2×10^{11} N/m ² .	4	L3	2	1, 2

Unit – II

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| 4. | a) | Discuss spontaneous emission and stimulated emission of radiation. | 4 | L2 | 3 | 1, 2 |
| | b) | Explain the construction and working of a Nd:YAG laser. | 8 | L2 | 3 | 1, 2 |
| | c) | Find the ratio of population of the two energy states at 300 K, the transition between which emits photons of wavelength 694.3 nm. | 4 | L3 | 3 | 1, 2 |
| 5. | a) | Explain the terms Metastable state and Population inversion. | 4 | L2 | 3 | 1, 2 |
| | b) | Define the terms numerical aperture and angle of acceptance. Derive an expression for numerical aperture of an optical fiber in terms of R.I. of core and cladding. | 8 | L2 | 4 | 1, 2 |
| | c) | A continuous wave laser emits light at a wavelength of 632.8 nm and has an output power of 5 mW. How many photons are emitted in each minute by this laser when operated? | 4 | L3 | 3 | 1, 2 |
| 6. | a) | Describe an optical fiber? Explain the principle based on which optical transmission is achieved through a fiber? | 6 | L2 | 4 | 1, 2 |
| | b) | Explain with suitable sketches the different step index fibers. | 6 | L2 | 4 | 1, 2 |
| | c) | The refractive index of the core and cladding are 1.48 and 1.45 respectively. Calculate the acceptance angle of the fiber. If the diameter of the fiber is 60 μm , find the number of modes the fiber can support at an operating wavelength of 1.5 μm . | 4 | L3 | 4 | 1, 2 |

Unit – III

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|----|----|--|---|----|---|------|
| 7. | a) | Derive Sabine's formula for reverberation time. | 6 | L2 | 5 | 1, 2 |
| | b) | Write a note on impact of noise in multi-storied buildings. | 6 | L2 | 5 | 1, 2 |
| | c) | Distinguish between musical sound and noise. | 4 | L3 | 5 | 1, 2 |
| 8. | a) | Define reverberation time of a hall. Explain clearly what causes reverberation and how it can be minimized. | 6 | L2 | 5 | 1, 2 |
| | b) | Describe the classification of sound. What are the characteristics of musical sounds? | 6 | L2 | 5 | 1, 2 |
| | c) | A hall of volume 2265 m^3 with total absorption of 92.9 m^2 of open window. Calculate the effect on reverberation time if an audience fills the hall and the absorption increases by another 92.9 m^2 . | 4 | L3 | 5 | 1, 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
